

**NATIONAL UNIVERSITY OF MODERN
LANGUAGE ISLAMABAD**

Lab final



Submitted to:

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TASK 1

I

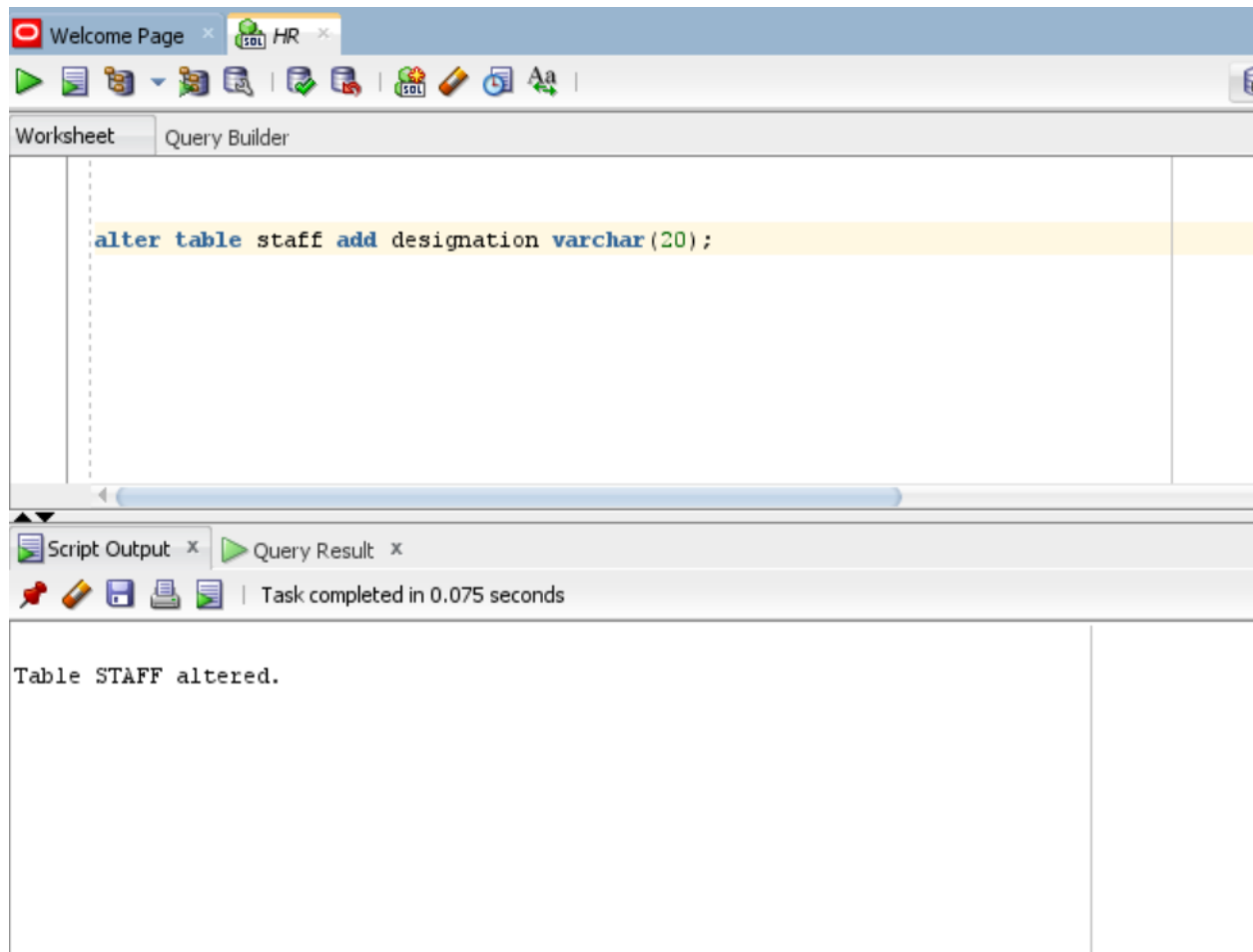
The screenshot displays a SQL IDE interface. At the top, there are tabs for 'Welcome Page' and 'HR'. Below the tabs is a toolbar with various icons for file operations, execution, and formatting. The main workspace is divided into two panes: 'Worksheet' and 'Query Builder'. The 'Worksheet' pane contains a SQL script to create a table named 'staff'. The script is as follows:

```
CREATE TABLE staff (  
    sid int PRIMARY KEY,  
    sname VARCHAR(20),  
    gender VARCHAR(10),  
    dept VARCHAR(20),  
    doj DATE  
);
```

Below the main workspace, there is a 'Script Output' pane and a 'Query Result' pane. The 'Script Output' pane shows the message 'Table STAFF created.' and a status bar at the bottom indicates 'Task completed in 0.104 seconds'.

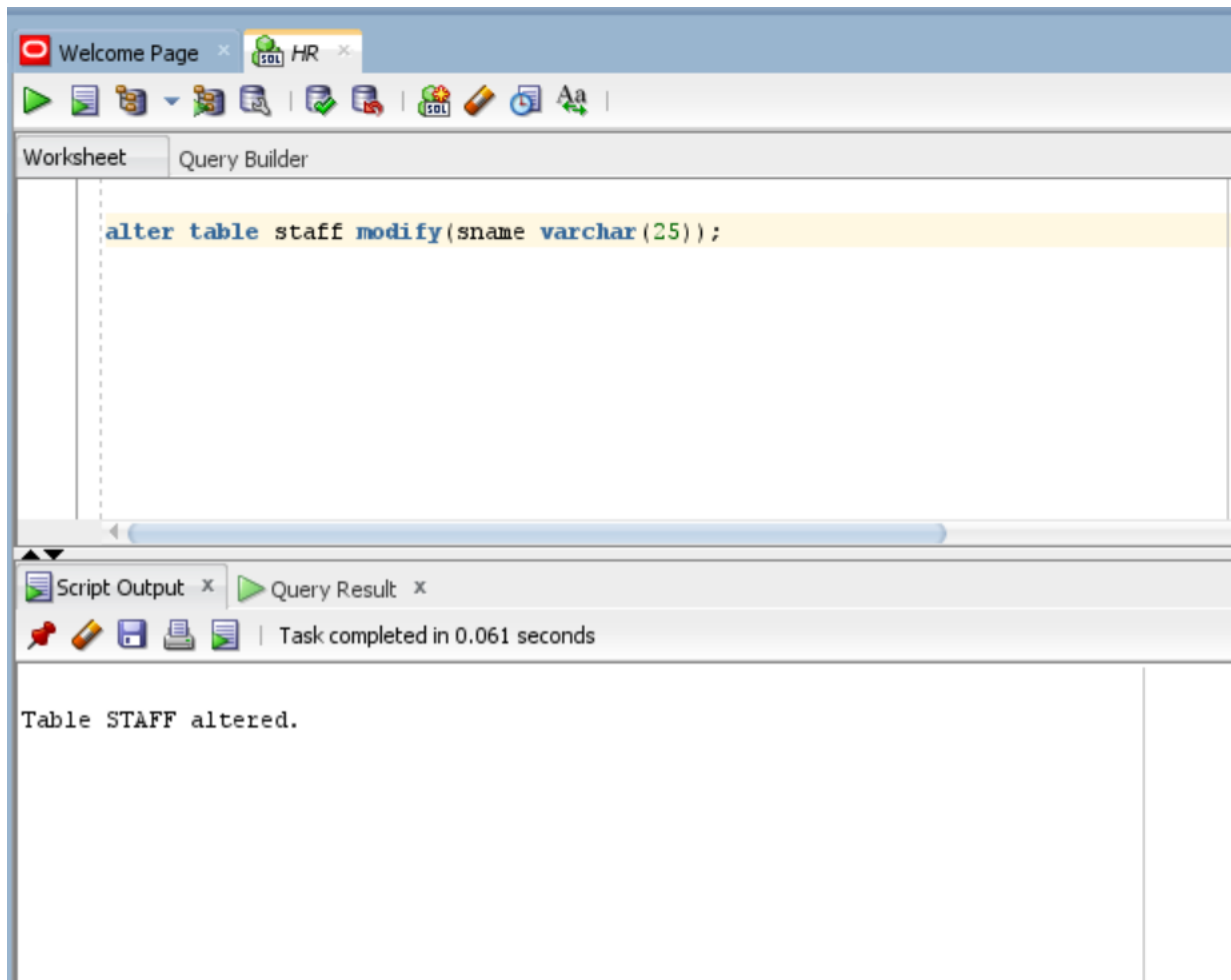
TASK 1

II



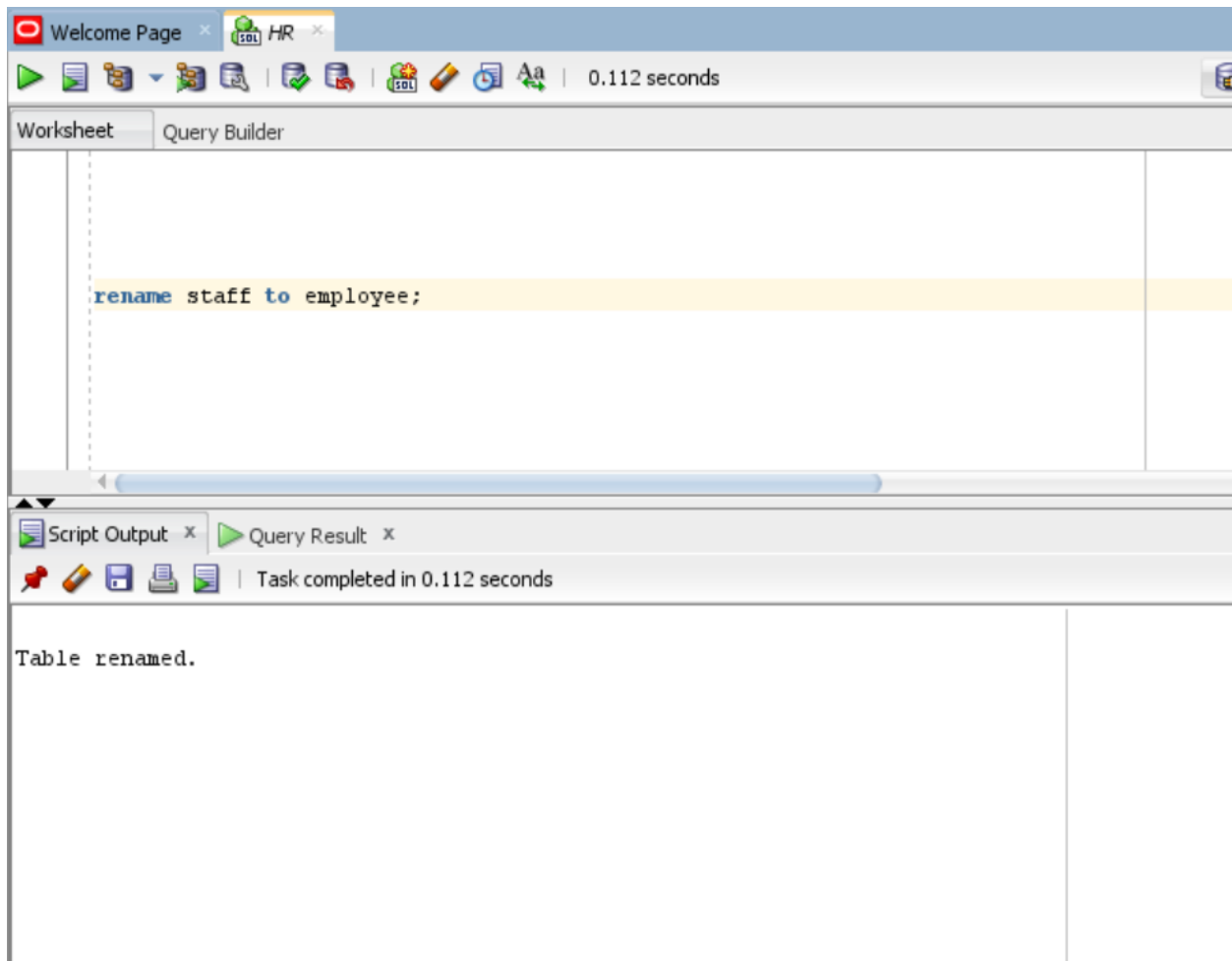
TASK 1

III



TASK 1

IV



TASK 1

V

The screenshot shows a database management tool interface. The top window, titled 'Welcome Page' and 'HR', contains a 'Query Builder' tab with a SQL script. The script consists of four 'INSERT INTO' statements for the 'employee' table, each adding a new record with specific values for 'sid', 'sname', 'gender', 'dept', and 'doj'. Below the script, a 'Script Output' window shows the results of the execution, indicating that each of the four rows was successfully inserted.

```
INSERT INTO employee (sid, sname, gender, dept, doj) VALUES ('001', 'John Smith', 'Male', 'Human Resources', TO_DATE('2020-03-15', 'YYYY-MM-DD'));
INSERT INTO employee (sid, sname, gender, dept, doj) VALUES ('002', 'Jane Doe', 'Female', 'Marketing', TO_DATE('2019-07-23', 'YYYY-MM-DD'));
INSERT INTO employee (sid, sname, gender, dept, doj) VALUES ('003', 'Emily Davis', 'Female', 'IT', TO_DATE('2021-01-10', 'YYYY-MM-DD'));
INSERT INTO employee (sid, sname, gender, dept, doj) VALUES ('004', 'Michael Brown', 'Male', 'Finance', TO_DATE('2018-11-01', 'YYYY-MM-DD'));
INSERT INTO employee (sid, sname, gender, dept, doj) VALUES ('005', 'Sarah Wilson', 'Female', 'Sales', TO_DATE('2022-05-19', 'YYYY-MM-DD'));
```

Script Output x

Task completed in 0.095 seconds

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

TASK 2

I

The screenshot displays a database management tool interface. At the top, there are two tabs: 'Welcome Page' and 'HR'. Below the tabs is a toolbar with various icons for file operations and execution. The main workspace is divided into two sections: 'Worksheet' and 'Query Builder'. The 'Query Builder' section contains a SQL query to create a table named 'Salary'.

```
CREATE TABLE Salary (  
    sid int,  
    scale VARCHAR2(50),  
    netsal NUMBER(10, 2),  
    CONSTRAINT fk_employee_sid FOREIGN KEY (sid) REFERENCES employee(sid)  
);
```

Below the query editor is a 'Script Output' section. It shows the result of the query execution: 'Table SALARY created.' and 'Task completed in 0.082 seconds'.

TASK 2

II

Worksheet Query Builder

```
desc employee;  
desc salary;
```

Script Output x

Task completed in 0.074 seconds

Name	Null?	Type

SID	NOT NULL	NUMBER(38)
SNAME		VARCHAR2(25)
GENDER		VARCHAR2(10)
DEPT		VARCHAR2(20)
DOJ		DATE
DESIGNATION		VARCHAR2(20)
Name Null? Type		

SID		NUMBER(38)
SCALE		VARCHAR2(50)
NETSAL		NUMBER(10,2)

TASK 2

III

The screenshot displays a SQL query execution environment. The top window, titled 'Welcome Page' and 'HR', shows a toolbar with various icons and a timer indicating '0.104 seconds'. Below the toolbar, the 'Query Builder' tab is active, displaying three SQL INSERT statements:

```
INSERT INTO Salary (sid, scale, netsal) VALUES ('001', 'A1', 60000);  
INSERT INTO Salary (sid, scale, netsal) VALUES ('002', 'B1', 50000);  
INSERT INTO Salary (sid, scale, netsal) VALUES ('003', 'C1', 70000);
```

Below the query editor, the 'Script Output' window is visible, showing the execution results:

```
1 row inserted.  
  
1 row inserted.  
  
1 row inserted.
```

TASK 3

I

The screenshot shows a SQL development environment. The top toolbar includes icons for running queries, saving, and editing. The 'Query Builder' tab is active, displaying the following SQL query:

```
SELECT AVG(netsal) AS average_salary  
FROM Salary;
```

Below the query editor, the 'Script Output' and 'Query Result' tabs are visible. The 'Query Result' tab shows the results of the query, indicating that all rows were fetched in 0.002 seconds. The results are displayed in a table with one column, 'AVERAGE_SALARY', and one row with the value 60000.

	AVERAGE_SALARY
1	60000

TASK 3

I

The screenshot shows the SQL Developer interface. The top toolbar includes icons for running queries, saving, and other database functions. The 'Worksheet' tab is active, displaying the following SQL query:

```
SELECT MAX(netsal) AS max_salary  
FROM Salary;
```

Below the query editor, the 'Query Result' tab is selected, showing the results of the query. The status bar indicates 'All Rows Fetched: 1 in 0.002 seconds'. The result is displayed in a table with one column, 'MAX_SALARY', and one row with the value 70000.

	MAX_SALARY
1	70000

TASK 3

II

The screenshot shows the SQL Developer interface, identical to the one above. The top toolbar includes icons for running queries, saving, and other database functions. The 'Worksheet' tab is active, displaying the following SQL query:

```
SELECT MAX(netsal) AS max_salary  
FROM Salary;
```

Below the query editor, the 'Query Result' tab is selected, showing the results of the query. The status bar indicates 'All Rows Fetched: 1 in 0.002 seconds'. The result is displayed in a table with one column, 'MAX_SALARY', and one row with the value 70000.

	MAX_SALARY
1	70000

TASK 3

III

The screenshot shows the SQL Developer interface. The 'Query Builder' tab is active, displaying the following SQL query:

```
SELECT MIN(netsal) AS min_salary  
FROM Salary;
```

Below the query editor, the 'Script Output' and 'Query Result' tabs are visible. The 'Query Result' tab shows the execution status: 'All Rows Fetched: 1 in 0.002 seconds'. The result set is displayed in a table with one column, 'MIN_SALARY', and one row with the value 50000.

	MIN_SALARY
1	50000

TASK 3

IV

The screenshot shows the SQL Developer interface. The 'Query Builder' tab is active, displaying the following SQL query:

```
SELECT COUNT(sid) AS number_of_employees  
FROM Salary;
```

Below the query editor, the 'Script Output' and 'Query Result' tabs are visible. The 'Query Result' tab shows the execution status: 'All Rows Fetched: 1 in 0.001 seconds'. The result set is displayed in a table with one column, 'NUMBER_OF_EMPLOYEES', and one row with the value 3.

	NUMBER_OF_EMPLOYEES
1	3

TASK 3

V

The screenshot shows a SQL query editor interface. The top bar includes a 'Welcome Page' tab and a 'HR' database icon. Below the toolbar, the 'Query Builder' tab is active, displaying the following SQL query:

```
SELECT SUM(netsal) AS total_salary  
FROM Salary;
```

Below the query editor, the 'Script Output' tab is active, showing the query results. The status bar indicates 'All Rows Fetched: 1 in 0.003 seconds'. The results are displayed in a table with one column, 'TOTAL_SALARY', and one row with the value '180000'.

	TOTAL_SALARY
1	180000

TASK 4

I

The screenshot shows a database query tool interface. The top bar includes a 'Welcome Page' tab and an 'HR' database tab. Below the top bar is a toolbar with various icons. The main area is divided into two tabs: 'Worksheet' and 'Query Builder'. The 'Query Builder' tab is active, displaying a SQL query: `select * from employee where (sname='Alishba' or sname='Ali');`. Below the query editor is a 'Query Result' tab, which shows the results of the query. The results are displayed in a table with 6 columns: SID, SNAME, GENDER, DEPT, DOJ, and DESIGNATION. There are 2 rows of data.

	SID	SNAME	GENDER	DEPT	DOJ	DESIGNATION
1	6	Alishba	Female	Finance	01-NOV-18	(null)
2	7	Ali	male	Sales	19-MAY-22	(null)

TASK 4

II

The screenshot shows a software interface with two main panes. The top pane, titled 'Query Builder', contains a text editor with the following SQL query: `select * from employee where sname like 'Y%';`. The bottom pane, titled 'Query Result', displays the results of the query. It shows a table with the following columns: `SID`, `SNAME`, `GENDER`, `DEPT`, `DOJ`, and `DESIGNA...`. The status bar indicates 'All Rows Fetched: 0 in 0.003 seconds'.

Query Builder

```
select * from employee where sname like 'Y%';
```

Query Result

All Rows Fetched: 0 in 0.003 seconds

SID	SNAME	GENDER	DEPT	DOJ	DESIGNA...
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TASK 4

III

The screenshot displays the Oracle SQL Developer application window. The top toolbar includes icons for running queries, saving, and other database functions. The main workspace is divided into two panes: "Worksheet" and "Query Builder". The "Query Builder" pane contains the following SQL query:

```
select * from employees where SALARY BETWEEN 25000 AND 50000;
```

Below the query editor is the "Query Result" pane. It shows the status "All Rows Fetched: 0 in 0.002 seconds". The result set is displayed as a table with the following columns:

EMPLOYEE...	FIRST_NA...	LAST_NAME	EMAIL	PHONE_N...	HIRE_DATE	JOB_ID	SA
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TASK 4

IV

The screenshot shows a software interface with a top toolbar containing various icons for file operations and execution. Below the toolbar, there are two tabs: 'Worksheet' and 'Query Builder'. The 'Query Builder' tab is active, displaying a SQL query in a text area: `select salary*12 annuual_salary from employees;`. Below the query editor, there is a 'Query Result' section. It includes a status bar indicating 'SQL | Fetched 50 rows in 0.003 seconds'. Below the status bar, a table displays the results of the query. The table has two columns: an index column and a column labeled 'ANNUAL_SALARY'. The results show 15 rows of data.

	ANNUAL_SALARY
1	288000
2	204000
3	204000
4	108000
5	72000
6	57600
7	57600
8	50400
9	144096
10	108000
11	98400
12	92400
13	93600
14	82800
15	132000

TASK 1

```
CREATE TABLE staff (
```

```
    sid int PRIMARY KEY,
```

```
    sname VARCHAR(20),
```

```
    gender VARCHAR(10),
```

```
    dept VARCHAR(20),
```

```
    doj DATE
```

```
);
```

```
alter table staff add designation varchar(20);
```

```
alter table staff modify(sname varchar(25));
```

```
rename staff to employee;
```

```
INSERT INTO employee (sid, sname, gender, dept, doj) VALUES ('001', 'John Smith', 'Male',  
'Human Resources', TO_DATE('2020-03-15', 'YYYY-MM-DD'));
```

```
INSERT INTO employee (sid, sname, gender, dept, doj) VALUES ('002', 'Jane Doe', 'Female',  
'Marketing', TO_DATE('2019-07-23', 'YYYY-MM-DD'));
```

```
INSERT INTO employee (sid, sname, gender, dept, doj) VALUES ('003', 'Emily Davis', 'Female',  
'IT', TO_DATE('2021-01-10', 'YYYY-MM-DD'));
```

```
INSERT INTO employee (sid, sname, gender, dept, doj) VALUES ('004', 'Michael Brown', 'Male',  
'Finance', TO_DATE('2018-11-01', 'YYYY-MM-DD'));
```

```
INSERT INTO employee (sid, sname, gender, dept, doj) VALUES ('005', 'Sarah Wilson', 'Female',  
'Sales', TO_DATE('2022-05-19', 'YYYY-MM-DD'));
```

```
INSERT INTO employee (sid, sname, gender, dept, doj) VALUES ('006', 'Alishba', 'Female',  
'Finance', TO_DATE('2018-11-01', 'YYYY-MM-DD'));
```

```
INSERT INTO employee (sid, sname, gender, dept, doj) VALUES ('007', 'Ali', 'male', 'Sales',  
TO_DATE('2022-05-19', 'YYYY-MM-DD'));
```

task 2

```
CREATE TABLE Salary (  
    sid int,  
    scale VARCHAR2(50),  
    netsal NUMBER(10, 2),  
    CONSTRAINT fk_employee_sid FOREIGN KEY (sid) REFERENCES employee(sid)  
);
```

```
desc employee;
```

```
desc salary;
```

```
INSERT INTO Salary (sid, scale, netsal) VALUES ('001', 'A1', 60000);
```

```
INSERT INTO Salary (sid, scale, netsal) VALUES ('002', 'B1', 50000);
```

```
INSERT INTO Salary (sid, scale, netsal) VALUES ('003', 'C1', 70000);
```

TASK3

```
SELECT SUM(netsal) AS total_salary FROM Salary;
```

```
SELECT AVG(netsal) AS average_salary FROM Salary;
```

```
SELECT MAX(netsal) AS max_salary FROM Salary;
```

```
SELECT MIN(netsal) AS min_salary FROM Salary;
```

```
SELECT COUNT(sid) AS number_of_employees FROM Salary;
```

TASK 4

```
select * from employee where (sname='Alishba' or sname='Ali');
```

```
select * from employee where sname like 'Y%';
```

```
select * from employees where SALARY BETWEEN 25000 AND 50000;
```

```
select salary*12 annuual_salary from employees;
```